

Delivery Ghost: Effects of Language Immersion and Interactivity in a Language Learning Game

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ABSTRACT

One of the most effective ways to learn a second language is to immerse oneself in it in the country where it is spoken. While this is not feasible for most language learners, language learning games may be able to replicate an immersive language learning experience. We developed *Delivery Ghost*, a web-based game to teach Mandarin Chinese to beginner-level learners, including complete beginners. In a 2x2 experiment ($N=159$), we tested how immersion and interactivity affect learning gains (pre-post test) and experiences (enjoyment, flow, perceived learning, cognitive load). Participants in all conditions showed significant learning gains even though no explicit, lecture-style teaching was provided. We also found evidence of substantial learning gains and positive learning experiences independent of the level of immersion (i.e., availability of hints in English) and interactivity (i.e., animation vs. gameplay). These results suggest that interactivity and immersion are less critical to learning at the beginner-level than a well-structured curriculum.

CCS CONCEPTS

• **Human-centered computing** → *Empirical studies in HCI*; • **Applied computing** → **Interactive learning environments**; **Distance learning**; **E-learning**; **Computer-assisted instruction**.

KEYWORDS

Language immersion, Language learning, Serious game, Language learning games, Foreign language learning

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1 INTRODUCTION

Immersive language learning is considered an effective and natural way to acquire a second language (L2). Here we define immersion as placing oneself in the language environment that the target language (L2) is spoken while limiting the use of the first language (L1). For example, emphasis on speaking L2 as part of a study abroad

experience has been shown to enhance vocabulary acquisition and use, oral fluency, and grammar acquisition [12, 14, 21, 22, 32, 33, 45]. However, immersive learning such as studying abroad and traveling to a country in which the target language is spoken is costly and not open to everyone. Alternatively, virtual environments emerge as new language learning environments [1, 4, 5, 10, 15, 44]. Games have received attention for their potential as a means of language learning [7, 17, 19, 30, 36, 39, 47]. The educational games literature suggests that games can introduce learning content via well-designed game mechanics (e.g., quests, levels, and reward systems) with rich interactivity using multimedia [25]. Games can provide not only language immersion but also make language learning immediately accessible to everyone in the world. Complete immersion (i.e., L2 only) can be a challenging environment depending on learners' language proficiency. So far, we have limited knowledge about (1) to what extent the game needs to be linguistically immersive and (2) how much interactivity is needed in virtual learning environment settings to learn languages.

We developed a game, *Delivery Ghost*, that teaches Mandarin Chinese to beginner-level learners in an linguistically immersive way. We leveraged game mechanics to teach vocabulary and expressions without lecturing L2 learning content in L1. Our contribution to the HCI community is that we examined the effect of language immersion and interactivity on learning outcomes and experiences specifically for beginner-level learners. We provide insights into the designs for language learning technologies that teach a foreign language (L2) in complete immersion (use L2 only) without hurting learning experiences and outcomes, with implications for making L2 learning scalable and open to any language speakers.

2 RELATED WORK

There are multiple games that are designed to teach L2 in a linguistically immersive manner. For example, Culbertson et al. [9] designed *Crystallize*, a collaborative language learning game that guides players to discover both the meaning of L2 words without direct teaching. Cheng et al. [8] further developed *Crystallize* with virtual reality (VR) extension and showed that participants felt more involved in a foreign culture. As another example, a commercial game, *Influent*, lets players explore in a 3D simulated environment and learn foreign vocabulary in a naturalistic way [23]. However, most of these games use L1 to provide explicit instructions or translations. This leads to an experience that is somewhat different from L2-only immersion. Thus, we investigated whether a game could approximate those circumstances more closely.

- Can games teach L2 to beginner-level learners without explicit instructions in L1?

The language education literature suggests that two factors are critical for successful learning in language immersion. First, the

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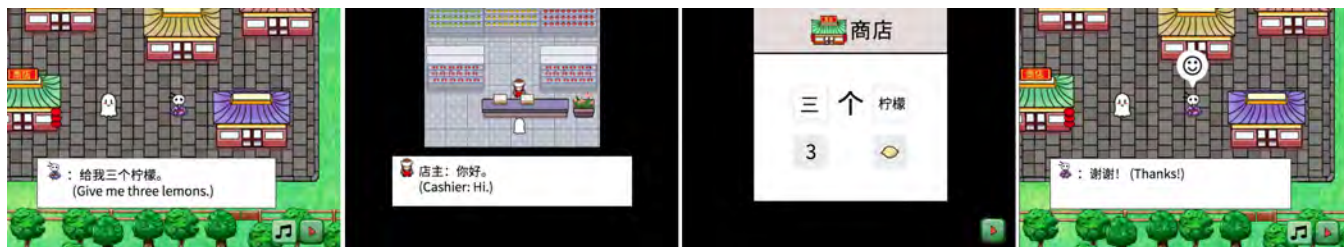


Figure 1: Selected screenshots of the animation in L1L2 in order (from left to right). Learners can click on the next button only and the fruit purchase and delivery will be shown with dialogues.

environment needs to provide learners with an adequate amount of contextualized language input in L1. According to Krashen’s scaffolding theory, language input must be comprehensible to learners for L2 proficiency improvement [27]. Comprehensible input refers to one-way input that is understandable yet one step further from the learner’s proficiency state. Exposing to L2 expressions in context helps make sense of L2, but the completely immersive language environment (i.e., only L2 is available) may not be comprehensible to beginner-level learners. Indeed, prior research on language immersion in class-room settings shows that teaching in L1 is necessary especially for students with a low-level L2 proficiency when introducing a complex grammar syntax [42]. Thus, we hypothesize that L1 translation is necessary to teach L2 in the online language learning environment.

- Beginner-level learners will learn more in the language immersion with L1 translation (L1L2) than in the complete language immersion (L2)
- Beginner-level learners will have more positive learning experience (flow, enjoyment, perceived learning, and cognitive load) in the language immersion with L1 translation (L1L2) than in the complete language immersion (L2).

Second, interaction with other L2 speakers, especially expert L2 speakers, is essential for L2 learning [26]. In the computer-assisted language learning environment, a computer act from the software system that engages with learners is also viewed as part of interaction [2]. Computer acts can also serve as scaffolding, which refers to support from a more knowledgeable person in the process of achieving a learning goal [46]. Games inherently incorporate a high level of interactivity with diverse scaffolding techniques, such as providing feedback, prompting next-level tasks, and directing to resources learners, removing provided assistance (i.e., fading). Beginner-level learners who require more scaffolding than any other proficiency levels [18, 26] may benefit more from games than other less interactive learning software programs. However, according to Cognitive load theory, interactivity adds cognitive load to learners and it may lower learning performance when available working memory is exhausted [35]. DeHaan and Kuwanda [11] found that playing a music video game in L2 was less effective in L2 vocabulary learning than watching someone playing the game, although the game was not purposefully designed for L2 learning. In sum, it is unclear that interactivity has a positive or negative impact on beginner-level learners’ language learning in the

learning materials that are designed for teaching L2. Thus, we pose the following research questions:

- How does the level of interactivity in the online learning materials (game vs. animation) affect beginner-level learners’ learning outcome?
- How does the level of interactivity in the online learning materials (game vs. animation) affect beginner-level learners’ learning experience (flow, enjoyment, perceived learning, and cognitive load)?

3 METHODS

3.1 Experiment Design

We conducted an online experiment using a 2-by-2 factorial design (Language immersion: complete immersion (L2) vs. translation-available condition (L1L2); Interactivity level: game vs. animation). The animation version was made with the same game assets used for the game, *Delivery Ghost*. The game more interactive than the animation; While in the game, learners control the player character to solve the quests, in animation learners can only click on a button to progress (Figure 1). The learning content was identical to the game condition, but the fading of visual aids in the game quests was not provided due to the absence of game quests. In the L1L2 conditions, English translation (L1) was available in the game, while the L2 conditions did not have any English words. The tutorials on the game rules were also provided using visual aids without L1 descriptions (Figure 2). We did not provide *pinyin*, the romanization system for Chinese, in any conditions, because we concerned that it may cause first language interference [40]. For example, representing the Japanese sound “r” with an English “r” confuses English speakers because the correct sound is different from an English “r.”

3.2 Delivery Ghost

We developed *Delivery Ghost*, a web-based game that teaches Mandarin Chinese to beginner-level learners, written in JavaScript using Phaser, an open-source HTML5 game framework. The game narrative is that a ghost shops for fruit for town residents, which create a reason for foreign language conversation in context. The learner maneuvers the ghost character to visit town residents and the store for fruit delivery. To replicate completely immersive language environment, explicit instructions are not included. Beginners thus learn Mandarin Chinese in an authentic context as expressions are introduced in context. For example, by repeatedly presenting



Figure 2: Example scenes of *Delivery Ghost* in L2 (first row) and L1L2 (second row): the game title, receiving a quest, and entering the store, and level selection scenes (from left to right). The L2 version does not have any English (L1) input, whereas the L1L2 version provides English (L1) translation.

a certain expression at the end of conversations, learners become aware of what they need to say when ending a conversation. This way, they can pick up vocabulary in a naturalistic way without lectures. We conducted two pilot tests to optimize learning tasks and improve the user experience for beginner-level learners, including game tutorials, length of the game, etc. The final version has 16 delivery quests in five levels. It takes approximately ten minutes to complete.

The learning objective was to understand simple conversational expressions (hi, bye, thank you, excuse me), five fruit names, numbers (1 - 3), and the basic syntax of Mandarin Chinese (noun phrase: number + modifier + noun, sentence: verb + direct object + indirect object)(Figure 3). Levels and Quests were designed to achieve this goal: A level consisted of 2 to 5 fruit delivery quests. The number of quests increases as the level goes up. A quest is to receive a delivery request from a town resident, purchase fruits at the store, and deliver the fruits to the town resident. Each step needs to engage in dialogue with non-player characters (NPCs). Learners were given options in the prompt, including “hi” and “bye” to initiate or end a conversation. An NPC gave a quest in conversation, saying, “Give me [number] [fruit],” and this request appears on the quest log immediately (refer to the screenshots in the second column Figure 2). When purchasing fruits at the store and giving the fruits to the town resident teaches not only vocabulary but also the syntax of noun phrases and sentences in Mandarin Chinese (Figure 3). Learners were continuously exposed to both Chinese characters and audio files for sentences, phrases, and words throughout the game.

To scaffold the learning process, we used fading and feedback. First, in order to teach the syntax of Mandarin Chinese smoothly, visual aids were gradually replaced with Mandarin Chinese in the upper levels (example can be found in Figure 3. Second, we provide feedback on the completion of quests and levels. When fruits were incorrectly delivered, a warning sound was played and the town resident said “excuse me” with an unhappy emoji. On the other hand, when fruits were correctly delivered, reward money was given with

coin sound and the town resident said: “thank you.” Additionally, when a level was complete, a festive firework animation was played and a star badge added next to the level button on the level selection scene (refer to the screenshots in the last column Figure 2).

3.3 Data Collection, Participants, and Study Procedure

University students were recruited from a research participation system at a large university in the US. A total of 159 students participated in the study for credit. The experiment was conducted individually and remotely through a study website. Participants were randomly assigned to one of the four conditions when entering into the study website. The study procedure was as follows: (1) background survey, (2) pre-test for learning content, (3) learning content based on randomly assigned condition, (4) post-survey, and (5) post-test for learning content. The study took about 20 minutes to finish.

We excluded those who reported that they can either read or understand at least a few phrases in Mandarin Chinese from analysis as the learning materials were designed to teach beginner-level learners. The final sample included 74 students (Game in L1L2: 17, Game in L2: 16, Animation in L1L2: 22, Animation in L2: 19).

3.4 Measures

The background survey assessed reading and listening proficiency in Mandarin Chinese with two questions (“How much do you understand Mandarin Chinese when it is spoken to you?” “How much do you read Mandarin Chinese?”) on a 5-point scale (0: Not at all, 1: A little, 2: A moderate amount, 3: A lot, and 4: A great deal). Participants’ demographic information was not collected.

Learning outcomes were measured by the pre-test and the post-test. The pre-test and the post-test asked the same questions to examine participants’ learning gains after the learning session. We created the tests reflecting the learning objectives. The test comprised 10 vocabulary questions (short answer type; writing



Figure 3: Left: Screenshots of purchasing fruits at the store (first screenshot) and giving fruits to a town resident from the inventory (second screenshot). Right: Examples of fading of quests in the quest log.

the meaning of given Chinese words and expressions), 3 syntax questions (multiple choice type; selecting the sentence written in Mandarin with the correct grammatical order), and 1 translation question (short answer type; writing translation of the given Chinese sentence) (see Figure 4 as examples). For the vocabulary and translation questions, audio files were also provided. Each question carries 1 point. A participant can earn up to 14 points in each test.

The post-survey examined participants' experience and attitude toward the learning materials they had. Survey constructs used in prior game-based learning research [3, 6] were adapted to measure flow, enjoyment, perceived learning, and cognitive load, and they were asked on a 5-point scale (0: Not at all, 1: A little, 2: A moderate amount, 3: A lot, and 4: A great deal). The survey items and Cronbach's α are the following: Flow (0.83; I lost track of time when I played the game; I really got into the game; I was very involved in the game; When I played I did not think of anything else), Enjoyment (0.90; I enjoyed the game; I had fun playing the game; Playing the game was pleasant), Perceived learning (0.83; I learned a lot from the game; The game added to my knowledge; I learned new things from the game), and Cognitive load (0.82; The learning content is very difficult; learning content is very complex; The learning content is not easy to be comprehended). We used the term, 'learning material' for animation condition instead of using 'game.'

3.5 Data Analysis

We use a two-way ANOVA to analyze the experimental data given the 2x2 factorial design. However, frequentist statistics cannot provide evidence for the null hypothesis of equivalence between two conditions (e.g., Do the game and the animation produce comparable learning outcomes?). We therefore additionally report results from Bayesian testing, which checks support for the null hypothesis [13, 29]. Bayes factor B quantifies the likelihood of the null hypothesis (H_0) that two sample means are equal, relative to the alternative hypothesis (H_1) that two sample means are different: if $B > 0$, there is support for H_1 , but if $B < 0$, there is support for H_0 [20]. We conducted Bayes factor testing using 'BayesFactor' R package [34] and followed Jeffreys' guide for interpretation [24]: anecdotal evidence ($B=1-3$ or $0.33-1$), substantial evidence ($B=3-10$ or $0.1-0.33$), strong evidence ($B=10-30$ or $0.03-0.1$), very strong evidence ($B=30-100$ or $0.01-0.03$), and decisive evidence ($B>100$ or <0.01) for H_1 (or inversely H_0).

4 RESULTS

At the start of the study, most of the participants in our final sample were unable to read or understand Mandarin Chinese at all:

only 17% indicated that they can read Mandarin Chinese a little, and 4% understood spoken Mandarin Chinese a little. Random assignment was successful, as there was no statistically significant difference between conditions in language proficiency prior to the study.

Beginner-level learners learned L2 without direct teaching. As shown in Figure 2, we found decisive evidence that participants learned during the study regardless of conditions ($B > 100$). On average, participants scored 8% in the pre-test (about 1 question answered correctly out of 14), but their score increased to 51% in the post-test (about 7 questions answered correctly) (Pre-test: $M = 1.19, SD = 4.1$; Post-test: $M = 7.15, SD = 2.1$). A t -test confirms the significant increase from pre- to post-test ($t_{73} = 12.92, p < 0.001$), but the increase did not significantly differ by condition ($F_{3,70} = 0.73, p = 0.54$).

L2 and L1L2 conditions resulted in comparable learning outcomes. Surprisingly, participants in L1L2 did not outperform those in L2. Instead, we found substantial evidence that the language environments with complete immersion (L2) and translation (L1L2) had comparable effects on learning gains ($B = 0.26$). In L1L2, participants scored 6.13 higher ($SD = 3.91$), about four times higher scores than their pre-test score. Similarly, in L2, participants scored 5.77 more in the post-test ($SD = 4.08$), about seven times higher than their pre-test score. A two-way ANOVA confirms that the L2 condition was not significantly different in learning outcomes from the L1L2 condition (main effect: $F_{1,70} = 0.99, p = 0.32$; interaction: $F_{1,70} = 2.76, p = 0.10$).

L2 and L1L2 conditions produced similar learning experiences. Surprisingly, participants in the L1L2 condition did not have more positive experiences than those in the L2 condition. We found substantial evidence that participants in the completely immersive environment (L2) had similar experiences as participants in the translation-available environment (L1L2) (Enjoyment: $B = 0.24$; Flow: $B = 0.24$; Perceived learning: $B = 0.27$; Cognitive load: $B = 0.30$). Both environments were perceived as moderately challenging (L2: $M = 2.10, SD < 0.01$; L1L2: $M = 1.92, SD < 0.01$). At the same time, both were also perceived moderately to very educational (L2: $M = 2.70, SD < 0.010$; L1L2: $M = 2.79, SD < 0.01$). Participants experienced flow a little to moderately (L2: $M = 1.78, SD < 0.010$; L1L2: $M = 1.78, SD < 0.01$) and they had a little to a moderate amount of fun (L2: $M = 1.95, SD < 0.01$; L1L2: $M = 1.93, SD < 0.01$). A two-way MANOVA confirms there was no significant difference between L1 and L1L2 in enjoyment, flow, perceived learning and cognitive load (main effect: $F_{4,70} = 0.17, p = 0.94$; interaction: $F_{4,70} = 0.51, p = 0.73$).



Figure 4: Vocabulary questions (left), a syntax question (middle), and a translation question (right).

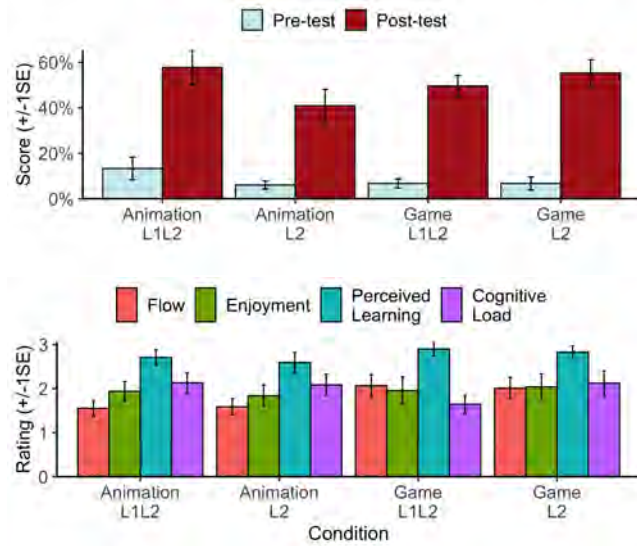


Figure 5: Pre-test vs. post-test scores between conditions (top). The test score was converted to a percentage. Self-reported learning experiences between conditions (bottom). The scale was from not at all (0) to a great deal (4).

Game and animation conditions resulted in comparable learning outcomes. We find substantial evidence that learning gains are comparable in the game and animation conditions ($B = 0.33$). On average, participants in the game scored 6.4 higher ($SD = 4.5$) in the post-test, which is equivalent to about seven times higher score than their average pre-test score. Participants in the animation scored 5.6 higher ($SD = 3.2$) in the post-test, about four times higher score than they earned in the pre-test score. A two-way ANOVA confirms that the game condition was not significantly different in learning outcomes from the animation condition (main effect: $F_{1,70} = 0.12, p = 0.73$; interaction: $F_{1,70} = 2.76, p = 0.10$).

Game and animation conditions produced mostly similar learning experiences. There was substantial evidence that participants enjoyed learning through the game and the animation ($B = 0.26$). A two-way ANOVA confirms that there was no significant difference in enjoyment between them (main effect: $F_{1,70} = 0.16, p = 0.69$; interaction: $F_{1,70} = 0.11, p = 0.74$).

Both provided a moderate amount to a lot of enjoyment (Game: $M = 2.0, SD = 1.21$; Animation: $M = 1.89, SD = 1.05$). We found anecdotal evidence that participants in the game were more likely to experience flow more than those in the animation ($B = 2.11$). A two-way ANOVA confirms the results, (main effect: $F_{1,70} = 5.00, p = 0.03$; interaction: $F_{1,70} = 0.05, p = 0.83$). While participants who played the game experienced a moderate amount to a lot of flow ($M = 2.04, SD = 0.99$), participants who interacted with the animated materials experienced a little to a moderate amount of flow ($M = 1.57, SD = 0.80$). Although it was also anecdotal, participants reported similar levels of challenge and perceived learning in the two learning modes (Cognitive load: $B = 0.36$, Game: $M = 1.88, SD = 1.0$; Animation: $M = 2.11, SD = 1.04$; Perceived learning: $B = 0.43$, Game: $M = 2.87, SD = 0.61$; Animation: $M = 2.66, SD = 0.9$). A two-way MANOVA confirms there was no significant difference between game and animation conditions in perceived learning and cognitive load (main effect: $F_{2,70} = 0.95, p = 0.39$; interaction: $F_{2,70} = 0.64, p = 0.53$).

5 DISCUSSION

We developed a new language learning game, *Delivery Ghost*, and conducted a randomized experiment with it to better understand how the level of immersion (L2 vs. L1L2) and the level of interactivity (game vs. animation) influence learning outcomes and experiences among novice learners. In all conditions, participants learned L2 without direct teaching. Surprisingly, participants did not outperform with translation and showed comparable learning gains in L2 and L1L2 conditions. They did not have more positive learning experiences with translation either. This implies a possibility of teaching L2 to any language speakers with one version of the software without localizing it (e.g., developing a French version for French speakers and German version for German speakers). It also means speakers of a minority L1 can have access to learn L2 without restrictions imposed by their L1.

We also found that the game provides comparable learning outcomes as an animation. The game was not more enjoyable than the animation, but it was more likely to provide a flow experience. Participants reported similar levels of perceived learning and cognitive load. We speculate that cognitive load from gameplay (familiarizing with game rules and maneuvering) may offset the benefits of more scaffolds via high interactivity in the game. This offset might have led to similar learning outcomes and participants' perception of difficulty and learning. By studying how the animation compares

to the game, we simplified the level of interactivity rather than adding a layer of more sophisticated interaction. In contrast, in the CHI community, most new work adds layers of interactivity (e.g., speech recognition [41] and augmented reality [16]). More research needs to be done with new technology to advance our understanding of the effect of increasing but also decreasing interactivity on learning outcomes, especially in the interplay with cognitive load and enjoyment.

Finally, learning science research suggests that games need to be designed with an evidence-based approach to achieve robust learning outcomes [31, 38, 43]. We call for more research on game designs reflecting learning science principles. Although we did not explain our game design process in detail to focus on our substantive research questions about learning modalities, we took many opportunities to integrate lessons from the learning sciences during the game design process. We speculate that adapting scaffolding techniques to our game design helped to make the L2 condition comparable to the L1L2 condition. In fact, learning science research suggests that scaffolding techniques should be carefully aligned with the learning goals [28, 37]. Future studies should investigate how to design games to reflect learning science principles and how these designs enhance learning outcomes.

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